

Euclid, Book I, Proposition I.2

To Place at a Given Point a Straight Line Equal to a Given Straight Line

Formal Reconstruction — Technical Documentation

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This note compares Euclid's original construction for Proposition I.2 with its reconstruction over the Hilbert foundation. In the CGJteamLab development the proposition follows immediately from Book Zero #49 (Layoff), illustrating the difference between the architectural roles of segment construction in Euclid's geometry and the Hilbert axiomatic framework.

1 Statement

Given a point A and a segment BC , construct a point D such that

$$AD \cong BC.$$

2 Euclid's Classical Construction

Let A be the prescribed endpoint and BC the given segment. Euclid constructs a segment beginning at A and equal to BC .



Figure 1: The data of Proposition I.2: a prescribed point A and a given segment BC .

Join A to B , and construct the equilateral triangle ABD on AB by Proposition I.1. Draw the circle centered at B through C , and produce DB until it meets that circle at G . Next draw the circle centered at D through G , and produce DA until it meets this second circle at L .

The proof can be summarized by

$$BC = BG, \quad DG = DB + BG, \quad DL = DA + AL,$$

together with

$$DB = DA, \quad DG = DL.$$

Subtracting the equal sides DB and DA from the equal radii DG and DL gives $BG = AL$. Since $BG = BC$, it follows that

$$AL = BC.$$

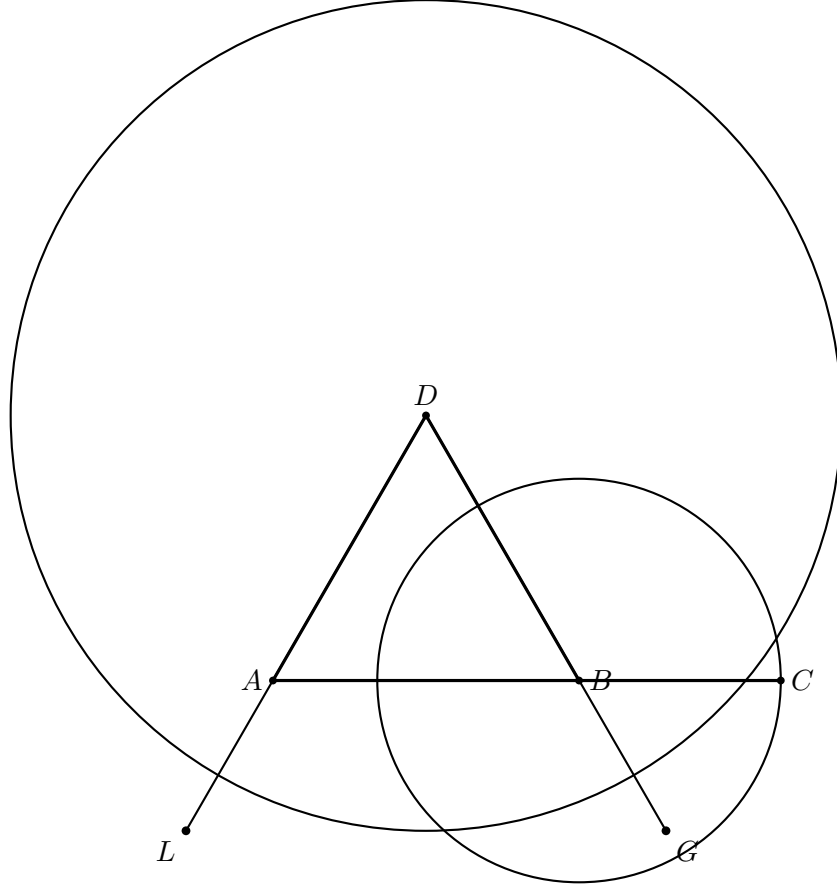


Figure 2: The complete classical construction. The first circle gives $BG = BC$; the second gives $DG = DL$. Since ABD is equilateral, $DB = DA$, and subtraction of equals yields $AL = BG = BC$.

Diagrammatic audit. The six successive construction snapshots in the previous version have been reduced to two figures: the initial data and the completed construction. The intermediate pictures added points and circles one at a time but did not represent distinct geometric cases.

Wilkins’s account also exposes a foundational subtlety that deserves mention. The sides DB and DA must be produced far enough to meet the relevant circles. This uses a strong reading of Euclid’s Second Postulate together with a tacit continuity or intersection principle. That issue belongs to the feasibility of Euclid’s construction, not to the Hilbert/Book Zero proof used below.

3 Hilbert Reconstruction

The formal theorem is not a literal transcription of Euclid’s construction. Its statement includes a chosen nondegenerate ray AE . The task is to place a copy of BC on that ray.

This is exactly Book Zero #49, `bookZero_49_layoff`. Given

$$A \neq E, \quad B \neq C,$$

the theorem produces a point D such that

$$\text{HilbertSameRay Geo A E D}$$

and

$$AD \cong BC.$$

Thus the equilateral triangle, both circles, the two produced sides, and the subtraction argument of Euclid disappear from Proposition I.2 itself. Their role is replaced by the already established abstract operation of laying off a segment on a prescribed ray.

4 Formalization

The complete implementation is available in the repository. In this case the mathematical content of the wrapper is literally one theorem call:

```
exact
  bookZero_49_layoff
    Geo A E B C hAE hBC
```

There is no additional incidence or congruence bookkeeping in `Proposition02.lean`. The two nondegeneracy assumptions are precisely the hypotheses required by the Book Zero layoff theorem.

5 Proof Architecture of the Reconstruction

```
Euclid
-----
point A + segment BC
      |
join AB
      |
I.1: equilateral ABD
      |
circle centered B through C
+ produce DB to G
      |
circle centered D through G
+ produce DA to L
      |
subtract equal segments
      |
AL = BC
```

```
Hilbert / Lean
-----
chosen ray AE + segment BC
      |
bookZero_49_layoff
      |
D on ray AE and AD ~= BC
      |
euclid_proposition_2
```

6 Discussion

Proposition I.2 is an especially clear example of compression by the Book Zero interface. Euclid devotes a nontrivial construction to transporting a length to a prescribed endpoint. The formal development treats that operation as an already established synthetic primitive.

This does not mean that the classical construction is dispensable mathematically. Rather, the proof burden has moved to the theorem `bookZero_49_layoff`. The Book I wrapper can therefore express the operation at the level at which later geometric proofs actually use it.

7 Relationship with Earlier Propositions

7.1 Proposition I.1

Proposition I.2 is the first substantial reuse of the equilateral triangle construction of I.1.

Euclid constructs equilateral triangles as auxiliary devices for transporting a given length to a prescribed point. The equilateral property creates rigid pairs of congruent segments from which the target segment congruence can be obtained.

Thus I.1 is not an isolated construction: its symmetry becomes working infrastructure immediately in I.2.

7.2 Transition to Proposition I.3

Proposition I.2 establishes free segment transport: given a segment and a prescribed point, a congruent segment can be constructed from that point.

Proposition I.3 adds an order constraint. When a longer segment is given, the transported length must be cut from inside it.

Hence the progression is

$$\text{I.2 : copy a length} \quad \longrightarrow \quad \text{I.3 : cut that length from a longer segment.}$$

8 Hilbert and Book Zero Components Used

8.1 Book Zero #49: Layoff

The sole substantive dependency of the formal proposition is `bookZero_49_layoff`. It packages existence of a point on a chosen ray together with congruence to a prescribed nondegenerate segment.

8.2 Same-Ray Structure

The conclusion records position by

HilbertSameRay Geo A E D.

This is stronger and more useful than merely asserting collinearity: it fixes the direction from A in which the copied segment is laid off.

8.3 Segment Congruence

The metric part of the conclusion remains purely synthetic:

$$AD \cong BC.$$

No numerical length function is introduced.

9 Formalization Notes

Proposition I.2 reveals an early difference between Euclid's constructional development and the Hilbert foundation.

Euclid has not yet postulated a general segment-layoff operation. Consequently he constructs the transported length indirectly using equilateral triangles and joins.

In Hilbert geometry, segment construction is already part of the congruence structure. The central mathematical content of I.2 is therefore available at a lower foundational level.

The dependency comparison is

Euclid: I.1 + auxiliary construction versus Hilbert: segment layoff.

This is analogous to what occurs much later in Proposition I.23, where Euclid derives angle transport through earlier triangle constructions while Hilbert III.4 provides angle construction directly.

Thus I.2 is the segment-level version of a recurring phenomenon in the reconstruction: a Euclidean construction theorem may become a thin interface theorem when the corresponding construction has been made foundational.

10 Dependency Summary

Classical:

I.1 + Postulates II, III
+ circle/line intersection
+ Common Notions

|
v

I.2

Formal:

bookZero_49_layoff

|
v

euclid_proposition_2

The formal theorem is therefore a thin Book I interface over a construction already established in Book Zero.

11 Conclusion

Proposition I.2 establishes the transport of segment length to a prescribed point.

For Euclid this is a genuine derived construction built from the equilateral-triangle machinery of I.1. In the Hilbert reconstruction, the corresponding segment-layoff principle belongs to the foundational congruence interface.

The contrast is mathematically informative. It shows, already at the second proposition of Book I, that the historical order of Euclid's constructions need not coincide with the dependency order of a modern axiomatic reconstruction.

I.2 immediately becomes infrastructure for I.3, where free transport is combined with order to cut a prescribed length from a longer segment. It is therefore the first general segment-construction interface of the Book I development.